Lecture 11 (*March 12*)

Reading: Section 4.1
Viewing: Lecture 14

Today we will start discussing orthogonality (i.e.perpendicularity). First of all, we know that two vectors $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$ are orthogonal precisely when their dot product is 0:

$$\boldsymbol{v} \perp \boldsymbol{w} \qquad \Leftrightarrow \qquad \boldsymbol{v} \cdot \boldsymbol{w} = \boldsymbol{v}^T \boldsymbol{w} = 0 \tag{88}$$

If two vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ are orthogonal, then we have the Pythagorean theorem:

$$||\boldsymbol{v} + \boldsymbol{w}||^2 = ||\boldsymbol{v}||^2 + ||\boldsymbol{w}||^2$$

where we recall formula (7) for the length $||\boldsymbol{v}||$ of a vector (proof of the Pythagorean theorem: $||\boldsymbol{v} + \boldsymbol{w}||^2 = (\boldsymbol{v} + \boldsymbol{w}) \cdot (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{v} \cdot \boldsymbol{v} + \boldsymbol{v} \cdot \boldsymbol{w} + \boldsymbol{w} \cdot \boldsymbol{v} + \boldsymbol{w} \cdot \boldsymbol{w} = ||\boldsymbol{v}||^2 + ||\boldsymbol{w}||^2$).

But now that we have the necessary language to talk about more than just individual vectors in $\mathbb{R}^n$, we can generalize the concept of orthogonality to entire subspaces:

Definition 11. *Two subspaces $V, W \subset \mathbb{R}^n$ are called* ***orthogonal****, which will be denoted by $V \perp W$, if we have $\boldsymbol{v} \perp \boldsymbol{w}$ for any two vectors $\boldsymbol{v} \in V$ and $\boldsymbol{w} \in W$.*

Note that two subspaces can be orthogonal only if $\dim V + \dim W \leq n$. Otherwise, the two subspaces would intersect in a non-zero vector $\boldsymbol{v} \in V \cap W$ (think about two planes passing through the origin in $\mathbb{R}^3$: they always contain a whole line) and it would be impossible for $\boldsymbol{v} \perp \boldsymbol{v}$.

Let us now consider an $m \times n$ matrix A, with the four fundamental subspaces considered in the previous Lecture. We have the following facts:

$$\boxed{C(A) \perp N(A^T)} \tag{89}$$

$$\boxed{N(A) \perp C(A^T)} \tag{90}$$

Let's prove these formulas: for (89), we need to take an arbitrary vector $\boldsymbol{a} \in C(A)$ and an arbitrary vector $\boldsymbol{v} \in N(A^T)$ and show that they are perpendicular. By the very definition of the column space and the left nullspace, we have:

$$\boldsymbol{a} = A\boldsymbol{w} \text{ for some vector } \boldsymbol{w} \qquad \text{and} \qquad A^T\boldsymbol{v} = 0 \;\Leftrightarrow\; \boldsymbol{v}^T A = 0$$

Then we have:

$$\boldsymbol{v} \cdot \boldsymbol{a} = \boldsymbol{v}^T \boldsymbol{a} = \boldsymbol{v}^T A\boldsymbol{w} = 0\boldsymbol{w} = 0 \qquad \Rightarrow \qquad \boxed{\boldsymbol{a} \perp \boldsymbol{v}}$$

Similarly, let's prove (90): we need to take an arbitrary $\boldsymbol{w} \in N(A)$ and an arbitrary $\boldsymbol{b} \in C(A^T)$ and show that they are perpendicular. By the very definition of nullspace and row space, we have:

$$A\boldsymbol{w} = 0 \qquad \text{and} \qquad \boldsymbol{b} = A^T\boldsymbol{v} \text{ for some vector } \boldsymbol{v}$$

Then we have:

$$\boldsymbol{b}^T = \boldsymbol{v}^T A \quad \Rightarrow \quad \boldsymbol{b}^T\boldsymbol{w} = \boldsymbol{v}^T A\boldsymbol{w} = \boldsymbol{v}^T 0 = 0 \quad \Rightarrow \quad \boxed{\boldsymbol{b} \perp \boldsymbol{w}}$$

(alternatively, you can think of statement (90) as simply restating (89) for A^T instead of A). Moreover, we saw in Fact 8 that the spaces in (89) are of complementary dimension, and thus they have the maximal total dimension that orthogonal subspaces can have. Ditto for (90). This is a special case of the following notion:

Definition 12. *Given a subspace $V \subset \mathbb{R}^n$, its* ***orthogonal complement*** *is the subspace $V^\perp$ consisting of all vectors in $\mathbb{R}^n$ perpendicular to V. In other words:*

$$V^\perp = \Big\{ \boldsymbol{w} \in \mathbb{R}^n \text{ such that } \boldsymbol{v} \perp \boldsymbol{w} \text{ for all } \boldsymbol{v} \in V \Big\} \tag{91}$$

Note that (91) is just a set. To show that it is a subspace, you would need to prove that:

$$\boldsymbol{v} \perp \boldsymbol{w}_1 \text{ and } \boldsymbol{v} \perp \boldsymbol{w}_2 \text{ for all } \boldsymbol{v} \in V \qquad \Rightarrow \qquad \boldsymbol{v} \perp (\alpha_1 \boldsymbol{w}_1 + \alpha_2 \boldsymbol{w}_2) \text{ for all } \boldsymbol{v} \in V$$

for any scalars α_1, α_2. The implication above can be easily proved by resorting to the dot product interpretation of perpendicularity. So combining Fact 8 with (89), (90) precisely says that:

the left nullspace is the orthogonal complement of the column space

the row space is the orthogonal complement of the nullspace

This is sometimes called the $\boxed{\text{Fundamental Theorem of Linear Algebra}}$. Let's see an example:

$$A = \begin{bmatrix} \boxed{1} & 0 & 3 & -1 \\ 0 & \boxed{1} & -2 & 1 \end{bmatrix}$$

(I chose a matrix which is already in reduced row echelon form to save us the work of Gauss-Jordan elimination, which is a routine computation for us by now). Then the row space of A is:

$$C(A^T) = \text{spanned by } \begin{bmatrix} 1 \\ 0 \\ 3 \\ -1 \end{bmatrix} \text{ and } \begin{bmatrix} 0 \\ 1 \\ -2 \\ 1 \end{bmatrix} \tag{92}$$

(we write the rows vertically to keep with our convention that vectors should be written as columns). To compute a basis for the nullspace of A, you must first note that the pivot columns of the matrix are 1 and 2, while the free columns are 3 and 4. Basis vectors for the nullspace are given by setting all the free variables equal to 0, except for one which is set equal to 1:

$$N(A) = \text{spanned by } \begin{bmatrix} a \\ b \\ 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} c \\ d \\ 0 \\ 1 \end{bmatrix}$$

To solve for a, b, c, d, you must use the property that the vectors above are annihilated by A:

$$A \begin{bmatrix} a \\ b \\ 1 \\ 0 \end{bmatrix} = A \begin{bmatrix} c \\ d \\ 0 \\ 1 \end{bmatrix} = 0 \quad \Rightarrow \quad \begin{cases} a + 3 = 0 \\ b - 2 = 0 \\ c - 1 = 0 \\ d + 1 = 0 \end{cases}$$

so we conclude that:

$$N(A) = \text{spanned by } \begin{bmatrix} -3 \\ 2 \\ 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \tag{93}$$

It is easy to see that the two basis vectors in (92) are orthogonal to each of the two basis vectors in (93), thus establishing the fact that the row space and nullspace are orthogonal subspaces.

The fundamental theorem of linear algebra says that the four fundamental subspaces of a matrix are pairwise orthogonal complements. Here's why we like this setup: if you have two complemetary subspaces $V, W \subset \mathbb{R}^n$ (as defined in Lecture 10) then any vector $\boldsymbol{a} \in \mathbb{R}^n$ can be written as:

$$\boldsymbol{a} = \boldsymbol{v} + \boldsymbol{w} \quad \text{where } \boldsymbol{v} \in V \text{ and } \boldsymbol{w} \in W \tag{94}$$

Moreover, $\boldsymbol{v}$ and $\boldsymbol{w}$ with this property are unique, and they are called the **components** of the vector $\boldsymbol{a}$ in the complementary subspaces V and W, respectively. So for example, letting V and W be the nullspace and row space of an $m \times n$ matrix, then we conclude that any vector in $\mathbb{R}^n$ can be written uniquely as a linear combination of the rows of A plus a vector annihilated by A.

In the special case when V, W are complementary <u>orthogonal</u> subspaces of $\mathbb{R}^n$, i.e.:

$$V = W^\perp \quad \text{or, equivalently} \quad W = V^\perp$$

the decomposition (94) takes on a more geometric meaning. In this case, $\boldsymbol{v}$ and $\boldsymbol{w}$ are none other than the projections of the general vector $\boldsymbol{a}$ onto the subspaces V and W:

$$\boldsymbol{v} = \text{proj}_V \boldsymbol{a} \qquad \text{and} \qquad \boldsymbol{w} = \text{proj}_W \boldsymbol{a}$$

We conclude that any vector is the sum of its projections onto complementary orthogonal subspaces.